

Aim:- To find the force constant of a helical spring by plotting graph b/w load and extension.

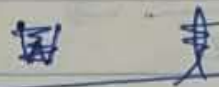
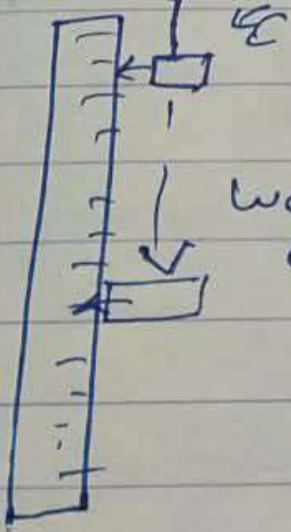
Acc. to Hooke's law:-

~~$F \propto kx$~~

$$F \propto l$$

$$F = kl$$

$$k = F/l \rightarrow \text{Spring Constant.}$$



$$W (\text{load on hanger}) = \text{Applied force } (F)$$

Reading of Position of Pointer End

W = F	l_u	l_v	$\frac{l_u + l_v}{2}$	(1)
250 N	29	23.5	23.25	
500 N	34.5	35		
750 N	42	43		
1000 N	49	44		
1250 N	54	45		

$$\frac{34.5 + 35}{2} \rightarrow 23.5 + 23 \rightarrow 23.25$$

$$\frac{69.5}{2} \rightarrow 250 \rightarrow \frac{29 + 29}{2} \rightarrow 29$$

$$\frac{104}{2} \rightarrow 500 \rightarrow \frac{34.5 + 35}{2} \rightarrow 34.75$$

$$\frac{146}{2} \rightarrow 750 \rightarrow 42$$

$$\rightarrow 49$$

$$\rightarrow 45$$

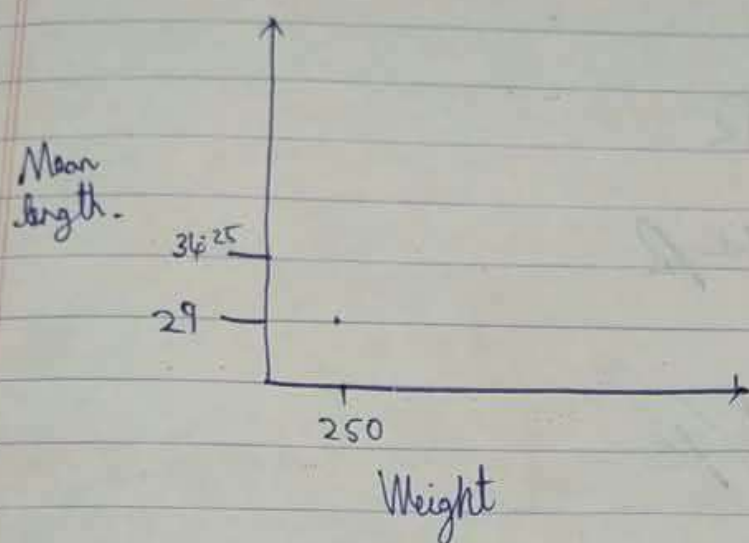
23.25

$$m = \frac{41 - 42}{1000 - 750} = \frac{99 - 42}{250} = \frac{2}{250}$$

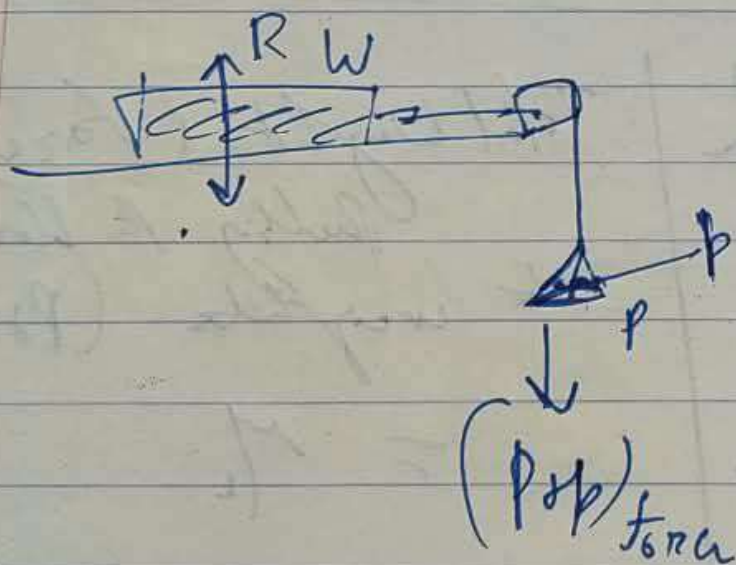
$$= \frac{7}{1/4} = 7 \times 4 = 28 = K$$

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AIM - To study the relationship b/w $\frac{F}{A}$ & N and to $\frac{y}{x}$ b/w block & the surface.



Measure Block weight
Measure force weight

add small weight until blocks start moving

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$$W = 100$$

$$P = 100$$

100	-	1
50	-	4
20	-	2
10	-	2
5	-	4

R
W + W (N)

F
P + P (F)

~~100~~

~~100~~

①

100

$$45 + 10 = \$55$$

②

~~150~~ 200

110

③

~~150~~ 150

~~80~~ 80

④

120

50

⑤

200

140

$$y_1 = 55/100 = 11/20$$

$$y_2 = 110/200 = 11/20$$

$$y_3 = 80/150 = 8/15$$

$$y_4 = 50/100 = 5/10 = 1/2$$

$$y_5 = 140/200 = 7/10$$

$$F \propto R$$

$$F = \mu R$$

$$\mu = F/R$$

S No.	Weight of wooden Block (w)	Total weight Pulled (w + w) = N
1.	0	100 + 0 = 100

weight on (P)	Total Weight (force) Pulling the block + weights (P & w)
0	
45 + 10 = 55.	= μ

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$$\frac{11}{20} + \frac{11}{20} + \frac{8}{15} + \frac{5}{12} + \frac{7}{15}$$

$$\frac{22}{20} + \frac{\cancel{180} \cancel{967} 75}{180} + \frac{7}{1}$$

$$\frac{22}{20} + \frac{15}{15} + \frac{5}{12}$$

$$\frac{11}{20} + \frac{5}{12} = \frac{33 + 100}{20(12)} = \frac{232}{240}$$

$$\frac{232 + 240}{240} = \frac{472}{240}$$

$$\frac{59}{20} = 59 \frac{1}{20}$$

Expt. No. 1

Experiment 1

AIM:- To measure diameter of a small spherical/cylindrical body using Vernier Calliper.

APPARATUS:- Vernier Callipers, a spherical body (pendulum bob) or a cylinder and a magnifying lens.

THEORY:- If with the body b/w the jaws, the zero of the Vernier scale lies ahead of the n^{th} division of main scale, then main scale reading (M.S.R.) = N

If n^{th} division of Vernier scale coincides with any division of main scale, then Vernier scale reading (V.S.R.)

$$= n \times (L.C.) \quad (L.C. \text{ is least count of Vernier calliper})$$

$$= n \times (V.C.) \quad (V.C. \text{ is Vernier constant of Vernier calliper})$$

$$\text{Total Reading} = T.R = M.S.R + V.S.R \\ N + n(V.C.)$$

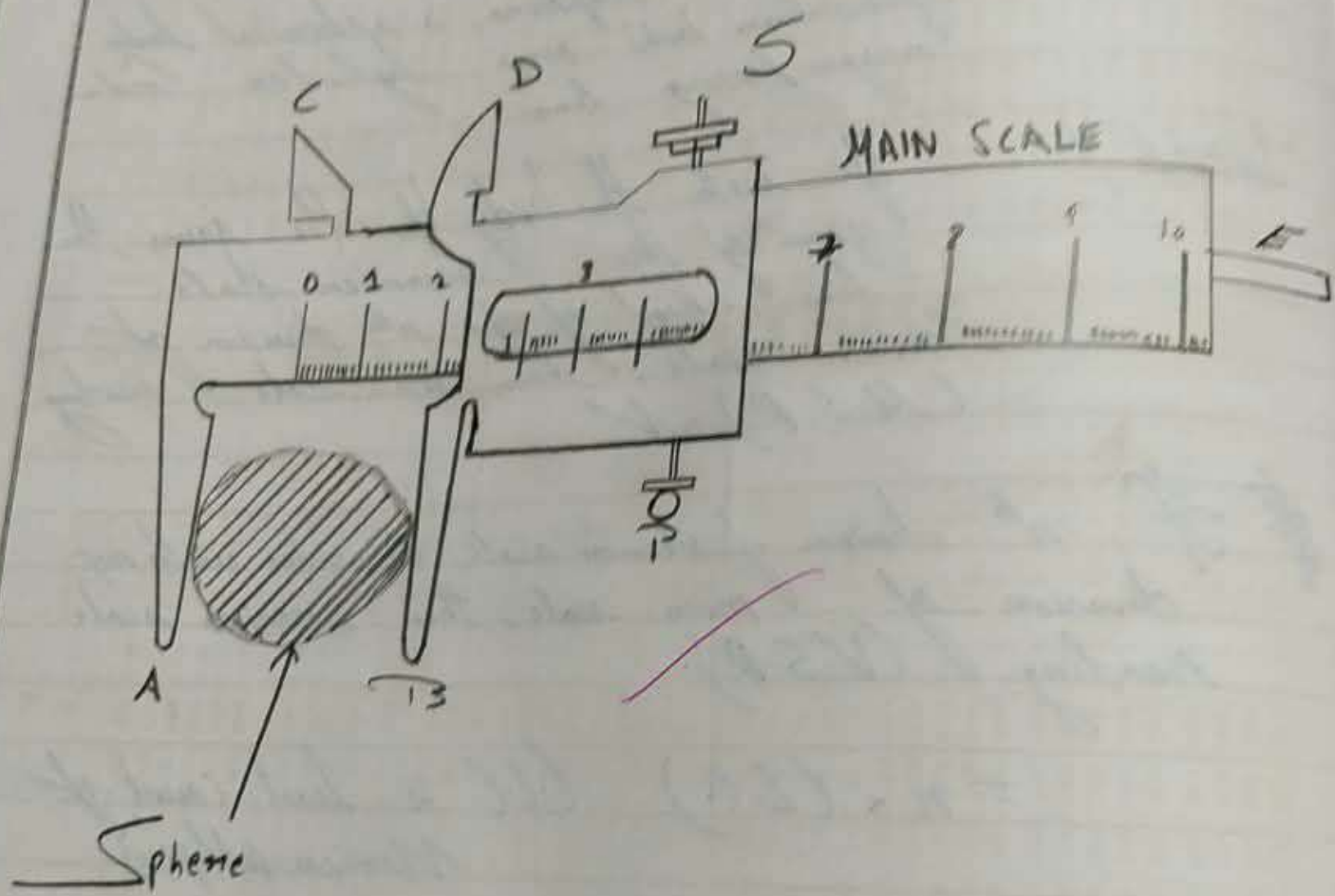
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PROCEDURE:-

1. Determine the Vernier constant (V.C.) i.e. least count (L.C.) of the Vernier callipers and record it stepwise.
2. Bring the movable jaw B.D in close contact with the fixed jaw A.C and find the zero error. Do it three times and record them. If there is no error, record zero error as nil.
3. Open the jaws, place the sphere on cylinder b/w the jaws A and B and adjust the jaw D.B. such that it gently grips the body w/o any undue pressure on it. Tighten the screws S attached to the Vernier scale V.
4. Note the number (n) of the Vernier scale division.
5. 4. Note the position of the zero mark of the Vernier scale on the main scale. Record the main scale reading just before the zero marks of the Vernier scale. This reading (N) is called main scale reading (M.S.R.)



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5. Note the number (n) of the Vernier scale division which coincides with some divisions of the main scale.
6. Repeat step 4 and 5 after rotating the body by 90° for measuring the diameter in a perpendicular direction.
7. Repeat steps 3, 4, 5 and 6 for three different positions. Record the observations in each set in a tabular form.
8. Find total reading and apply zero correction.
9. Take mean of different values of diameter and show that in the result with proper units.

RESULT

The diameter of the given sphere is ~~2.2~~ 1.65 cm.

Precautions :-

1. Motion of Vernier scale on main scale should be made smooth (by oiling, if necessary).



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2. Vernier constant and zero error should be carefully found and properly recorded.
3. The body should be gripped by the jaws firmly but gently.
4. Observations should be taken at right angles at one place and taken at least at three different places.

SOURCES OF ERROR:

1. The vernier scale may be loose on main scale.
2. The jaws may not be at right angles to the main scale.
3. The graduation graduations on scale may not be correct and clear.
4. Parallax may occur while taking observations.

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OBSERVATION

1. Determination of Vernier Constant (Least Count) of the Vernier Caliper.

$$1 \text{ MSD} = 1 \text{ mm}$$

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = \frac{9}{10} \text{ MSD} = 0.9 \text{ mm}$$

$$\text{Vernier Constant, V.C} = 1 \text{ MSD} - 1 \text{ VSD} = (1 - 0.9) \text{ mm}$$

$$= 0.1 \text{ mm}$$

$$= 0.01 \text{ cm}$$

$$\text{Zero Error (e)} = 0.4 \text{ cm}$$

$$\text{Correction (c)} = -e = -0.4 \text{ cm}$$

S. No.	MSR	VSD	Total Reading
		(n) (n x V.C)	
1	1.2	4 0.04	1.24
2	2.4	2 0.02	2.42
3	2.45	4 0.04	2.49

Calculation

$$D_o = \frac{1.24 + 2.42 + 2.49}{3} = \frac{6.15}{3} = 2.05 \text{ cm}$$

$$D_o = 2.05 - 0.4 = 1.65 \text{ cm}$$

$$r = \frac{D_o}{2} = 0.825 \text{ cm}$$

$$\text{Volume} \Rightarrow \frac{4}{3} \pi r^3$$

$$\approx 2.35 \text{ cm}^3$$

S.No.	M.S.R	V.S.D	D
1	1.3 cm	3	$1.3 + 0.03 = 1.33$
2	1.25 cm	1	$1.25 + 0.01 = 1.26$
3	1.2 cm	1	$1.2 + 0.01 = 1.21$

$$D_o = \frac{1.33 + 1.26 + 1.21}{3} = \frac{3.79}{3} = 1.263 \text{ cm}$$

$$D = D_o - c \Rightarrow 1.263 - ~~0.4~~ 0.4 = 0.863 \text{ cm}$$

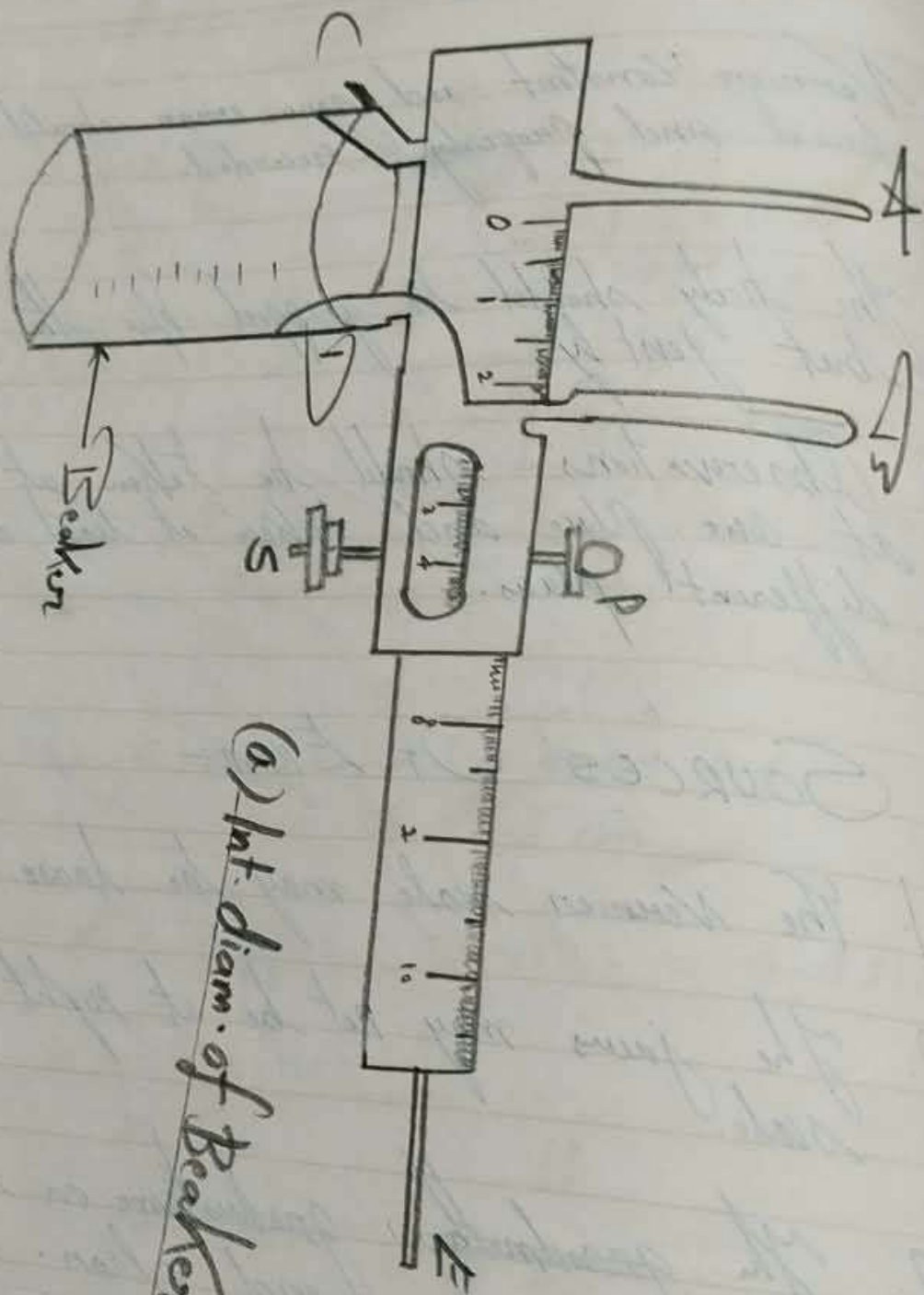
$$r = \frac{D}{2} = \frac{0.863}{2} = 0.4316 \text{ cm}$$

S.No.	M.S.R	V.S.D	H
1	3.7 cm	8	3.78 cm
2	3.75 cm	8	3.83 cm
3	3.9 cm	4	3.94 cm

$$H_o = \frac{3.78 + 3.83 + 3.94}{3} = \frac{11.55}{3} = 3.85 \text{ cm}$$

$$H - c = 3.45 \text{ cm}$$

$$V = \pi r^2 h = 3.14 \times 0.4 \times 3.85 \approx 2.25 \text{ cm}^3$$



(a) Int. diam. of Beaker

EXPERIMENT-2

AIM:- To measure internal diameter and depth of a given beaker/calorimeter using vernier callipers and hence find its volume.

APPARATUS:- Vernier Callipers, a beaker or a calorimeter, magnifying glass.

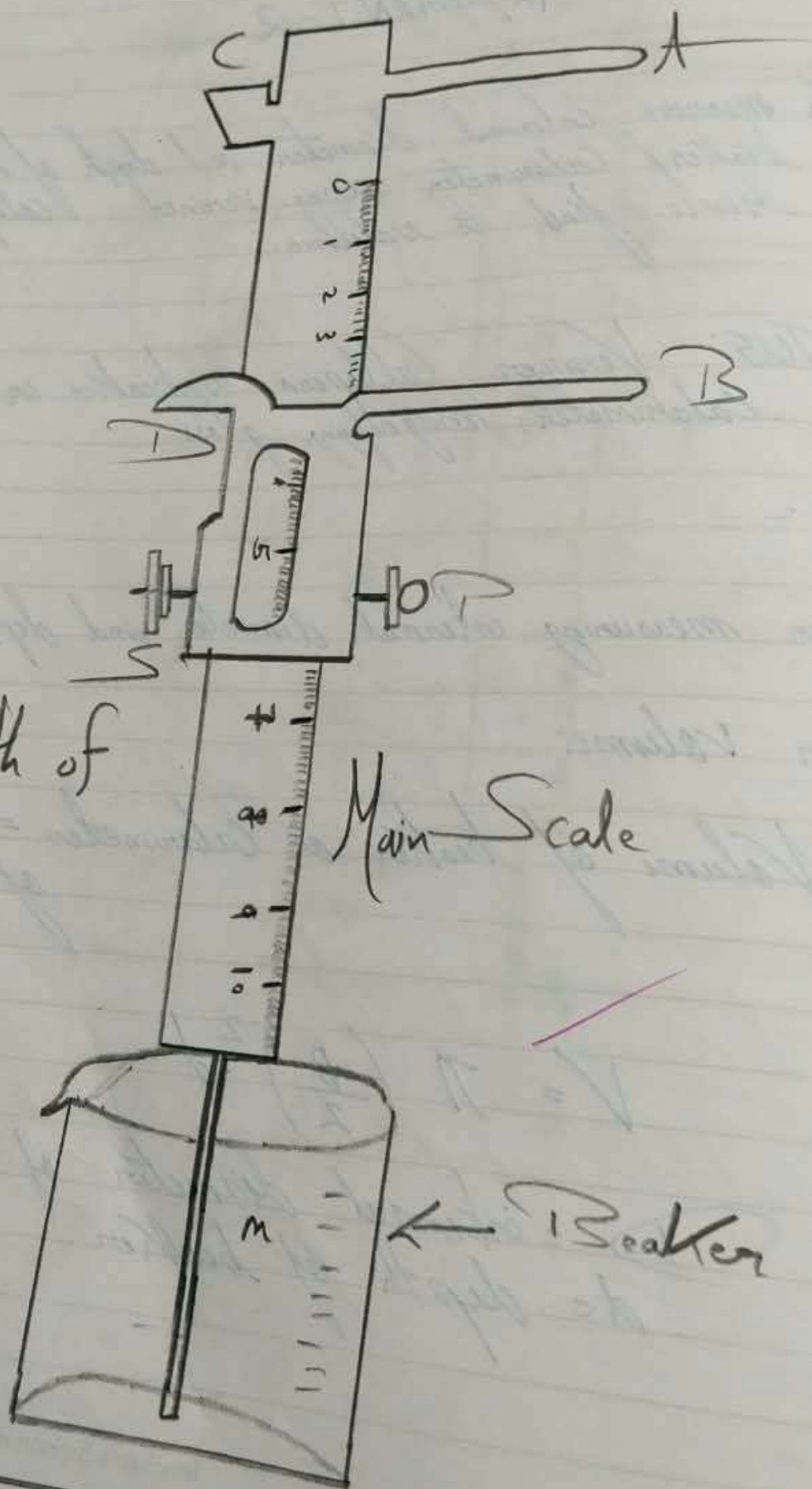
THEORY:-

- i) For measuring internal diameter and depth.
- ii) For volume:

Volume of beaker or calorimeter = internal area of cross section $\times$ depth

$$V = \pi \left(\frac{D}{2} \right)^2 \cdot d$$

D = internal diameter of beaker
 d = depth of beaker



(b) Depth of
Beaker

Main Scale

Beaker

PROCEDURES:

1. Determine the vernier constant (V.C.) i.e., least count of the vernier calipers and record it stepwise.
2. Bring the movable jaw B in close contact w/ the fixed jaw A and find the zero error. Do it three times and record it. If there is no zero error the second, zero error is nil.

Measurement of Internal diameter

3. Put the jaws C and D inside the beaker and open them till each of them touches the inner wall of the beaker, w/o any undue pressure on the walls. Tighten the screw attached to the vernier scale gently.
4. Note the position of the zero mark of the vernier scale on main scale. Record the main scale reading just before the zero mark of the vernier scale. This reading (V) is called main scale reading (M.S.R.).



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OBSERVATIONS:

1. Determination of Vernier Constant (L.C.) of the vernier callipers.

$$1 \text{ MSD} = 1 \text{ mm}$$

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ VSD} = 9/10 = 0.9 \text{ mm (M.S.D)}$$

$$\text{V.C.} = 1 \text{ MSD} - 1 \text{ VSD} = (1 - 0.9) \text{ mm} = 0.1 \text{ mm}$$

2. Zero Error

$$\text{Zero Error} = 0.4 \text{ cm}$$

$$\text{Zero Correction} = -0.4 \text{ cm}$$

3. Int. Diameter:-

S.No.	MSR	VSD	D
1	4.9 cm	2	4.92 cm
2	5.0 cm	2	5.02 cm
3	5.1 cm	3	5.13 cm

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- upper edge of the beaker or calipers.
11. Find total reading and apply zero correction.
 12. Take mean of two different values of internal diameter and two different values of depth.
 13. Calculate the volume by using proper formula and show that in result is proper unit.

RESULT

The volume of the beaker is 124.8287 cm^3 .

PRECAUTIONS:-

1. Motion of vernier scale on main scale should be made smooth (by oiling, if necessary).
2. Vernier constant and zero error should be carefully found and properly recorded.
3. The body should be gripped b/w the jaws firmly but gently.



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Expt. No. 2

5. Note the number (n) of the vernier scale division which coincides with some divisions of the main scale.
6. Repeat step 4 and 5 rotating the vernier callipers by 90° for measuring internal diameter in a perpendicular direction.
7. Find total reading and apply zero correction.

Measurement of depth:

8. Keep the edge of the main scale of vernier callipers on its peripheral edge. This should be done in such a way that the tip of the strip is able to go freely inside the beaker along its depth.
9. Keep sliding the moving jaw of the vernier callipers until the strip just touches the bottom of the beaker. Take care that it is just perpendicular to the bottom surface. Now, tighten the screw of vernier callipers.
10. Repeat steps 4 and 5 for four different position along the circumference of the

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$$D_o = \frac{4.92 + 5.02 + 5.13}{3}$$

$$= \frac{15.07}{3} = 5.023 \text{ cm}$$

$$D = D_o - \text{Zero Error} = 5.023 - 0.4$$

$$= 4.623 \text{ cm}$$

$$r = \frac{D}{2} = \frac{4.623}{2} \approx 2.3116 \text{ cm}$$

4. Depth :-

S.No.	M.S.R	VSD	H
1	7.3 cm	5	7.35 cm
2	7.34 cm	6	7.4 cm
3	7.5 cm	6	7.56 cm

$$H_o = \frac{7.35 + 7.4 + 7.56}{3} = \frac{22.31}{3}$$

$$\text{Volume} = \pi r^2 h = 7.436 \text{ cm}$$

$$= \frac{22}{7} \times (2.3116)^2 \times 7.436$$

$$\approx 124.82875 \text{ cm}^3$$

Expt. No. 2

4. Observation should be taken at right angles at one place and taken at least at three different places.

SOURCES OF ERROR:-

1. The vernier scale may be loose on main scale.
2. The jaws may not be at right angles to the main scale.
3. The graduations on scale may not be correct and clear.
4. Parallel may be there in taking observations.

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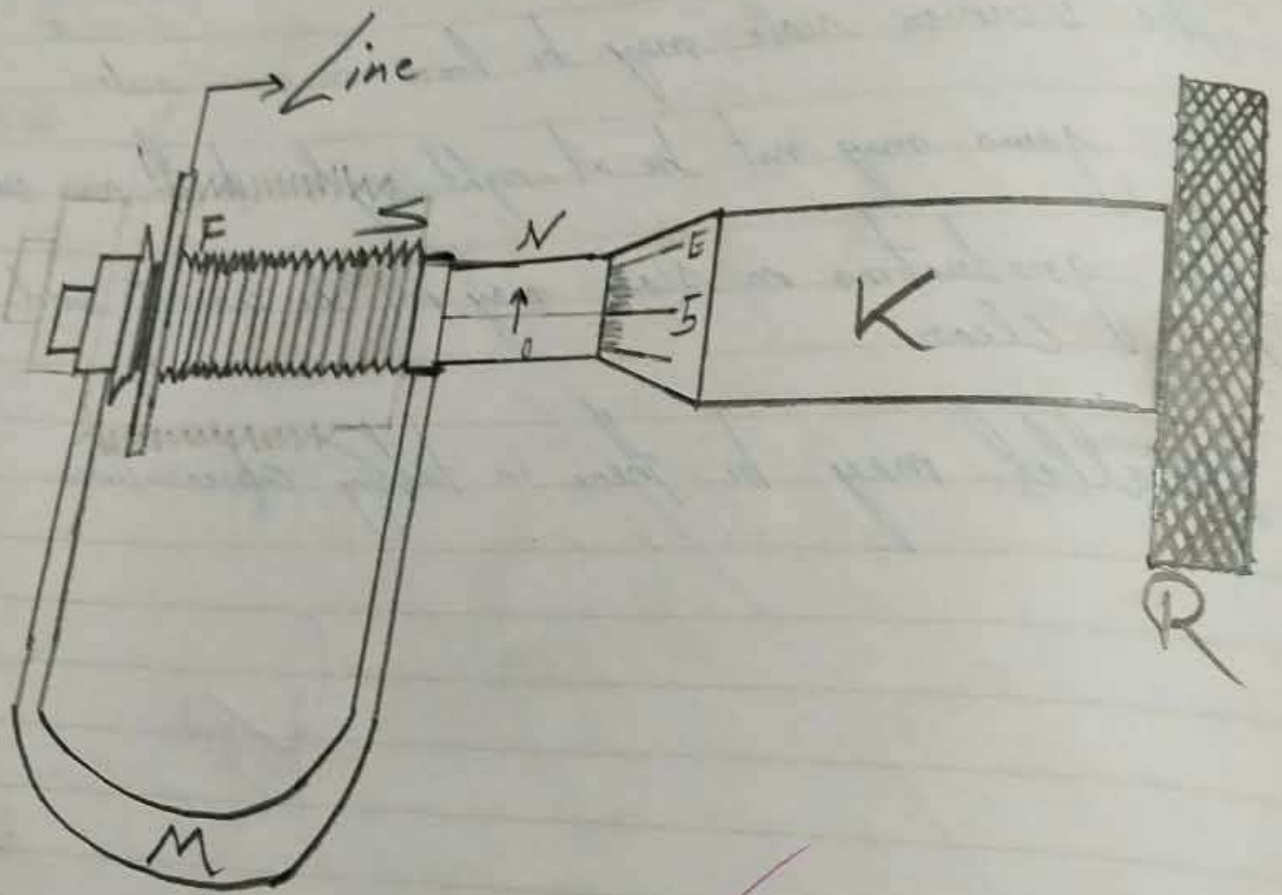


Fig - Screw Gauge measuring diameter of the wire.

EXPERIMENT-3

AIM:- To measure diameter of a given wire using screw gauge.

APPARATUS:- Screw gauge, wire, half meter scale and magnifying lens.

THEORY:-

1. If with the wire b/w plate faces A and B, the edge of the cap lies ahead of N^{th} division of linear scale, then, linear scale reading (L.S.R) = N .

~~If n^{th} division of circular scale lies ahead of N^{th} division of linear scale, then, linear scale reading (L.S.R) = N .~~

~~If n^{th} division of circular scale lies over reference line then, circular scale reading (C.S.R)~~

~~= $n \times (L.C.)$ (L.C is the least count of screw gauge)~~

$$\begin{aligned}\text{Total Reading (T.R.)} &= \text{L.S.R} + \text{C.S.R} \\ &= N + n \times (L.C.)\end{aligned}$$

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6. Note the number of the division of the divisions of the circular scale lying over reference line.
7. Repeat step 5 and 6 after rotating the wire by 90° for measuring diameter in a perpendicular direction.
8. Repeat steps 4, 5, 6 and 7 for five different positions separated equally throughout the length of the wire. Record the observation in each set in a tabular form.
9. Find total reading and apply zero correction in each case.
10. Take mean of different values of diameter.
11. Measure the length of the wire by stretching it along a half-meter scale. Keeping one end of wire at a known mark, note the position of the other end. Difference in position of two ends of wire gives the length of wire. Do it three times and record them.



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OBSERVATION:-

1. Determination of least count of the Screw Gauge

$$1 \text{ L.S.D} = 1 \text{ mm}$$

No. of full rotations given to screw = 4
Distance moved by screw = 4 mm

$$\text{Hence pitch } p = \frac{d}{h} = \frac{4 \text{ mm}}{4} = 1 \text{ mm}$$

Number of divisions on circular scale = 100

$$\text{Hence, least count} = \frac{1}{100} = 0.01 \text{ mm} = 0.001 \text{ cm}$$

2. Zero Error $\Rightarrow 0.8 \text{ mm} \Rightarrow -0.8 \text{ mm}$

3. S.No. M.S.R C.S.D D

1	0 cm	61	0.061 cm
2	0 cm	90	0.090 cm
3	0 cm	81	0.081 cm

Length of wire = 10 cm

$$P_0 = \frac{0.061 + 0.090 + 0.081}{3} = \frac{0.232}{3} = 0.0773 \text{ cm}$$

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2. If D be the mean diameter and l be the mean length of the wire,
 volume of the wire, $V = \pi \left(\frac{D}{2}\right)^2 l$

PROCEDURE:-

1. Find the value of one linear scale division (L.S.D).
2. Determine the pitch and the least count of the screw gauge and record it separately.
3. Bring the ~~plane~~ plane face B in contact w/ plane A and find the zero error. Do it three times and record them. If there is no zero error, then record zero error as nil.
4. Move the face B away from face A. Place the wire lengthwise over face A and move the face B towards face A using the ratchet head R. Stop when R turns (slips) w/o moving the screw.
5. Note the number of division of the linear scale visible and uncovered by the edge of the cap. The reading (N) is called linear scale reading (L.S.R).



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$$\therefore D_0 = 0.0773 \text{ cm}$$

$$\begin{aligned} D &= D_0 - \text{Zero Error} \\ &= 0.0773 - (-0.8) \\ &= 0.8773 \text{ cm} \end{aligned}$$

$$\text{Radius} = 0.43865 \text{ cm}$$

Volume of wire is

$$V = \pi \left(\frac{D}{2} \right)^2 l$$

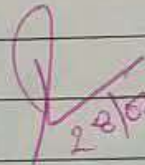
$$= \frac{22}{7} \times (0.43865)^2 \times 10$$

$$\approx 5.988156 \text{ cm}^3 \approx 6 \text{ cm}^3$$

7. Error due to Parallax should be avoided.

SOURCES OF ERROR:-

1. The screw may have friction.
2. The screw gauge may have back-lash error.
3. Circular scale division may not be of equal size.
4. The wire may not be uniform.


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RESULT

The volume of the given wire is 0.015286 cm^3 .
 5.988156 cm^3 .

PRECAUTIONS

1. To avoid undue pressure, the screw should always be rotated by ratchet R and not by cap K.
2. The screw should move freely w/o friction.
3. The zero correction, w/ proper sign should be noted carefully and added algebraically.
4. For some set of observation, the screw should be moved in the same direction to avoid back-lash error of the screw.
5. At each place, the diameter of wire should be measured in two $\perp$ dir. and then the mean of two be taken.
6. Readings should be taken at least for five different peaks equally spaced along the whole length of the wire.

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Length of simple pendulum = length of thread +
length of hook of bob +
length of radius of bob.

i.e. $L = L' + h + r$

3. Time period of simple pendulum - Time taken by the bob of the simple pendulum to make one complete vibration, is called the time period of simple pendulum. It is represented by symbol ' T '.

The time period is given by the formula,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

$$T^2 = \frac{4\pi^2 L}{g}$$

$$L = \frac{gT^2}{4\pi^2}$$

Knowing the values of T and g , L can be calculated

PROCEDURES:-

1. Find the vernier constant and zero error of the vernier caliper and record it.
2. Determine the mean diameter of the spherical bob. Find the mean radius (r) of the bob.
3. Find the length ' h ' of the hook attached to the bob by meter scale and record it.
4. Take a cotton thread about 2 meter long and fix its one end with the hook.
5. Put ink marks M_1 , M_2 and M_3 on the thread as distances of 10 cm, 15 cm, 20 cm, etc. from the center of the gravity of the bob. These distances give effective length (l) of the simple pendulum.
6. Pass the thread through the two split parts of a cork with the thread coming out just from 10 cm mark. (Let).
7. Tight the two half cork pieces by the clamp.

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EXPERIMENT-4

AIM:- Using a simple pendulum, plot its $L-T^2$ graph and use it to find the effective length of 2nd Pendulum.

APPARATUS:- A clamp with stand, a split cork, thread, bob vernier callipers, stop clock/watch, meter scale and a piece of chalk.

THEORY:-

1. Simple Pendulum - An ideal simple pendulum consists of a heavy point mass (called bob) tied to one end of a perfectly inextensible, flexible and weightless string. There is no ideal simple pendulum. In practice, we make a simple pendulum by tying a metallic spherical bob to a fine cotton stitching thread.
2. Length of simple pendulum - The distance b/w the point of suspension of the pendulum and its C.G. (which is C.G. of the bob), is called the length of the simplest pendulum. It is represented by the symbol 'l'.

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15. In this way count upto 19. Become alert when bob start the 20th vibration.
16. Just when 20th vibration is completed, count 20 and at once stop the stopwatch.
17. Find total time noting positions of both the hands of the clock. This time is time for twenty vibration.
18. Repeat step 13 to 18 two times more for same length.
19. Move the clamp up by 10 cm.
20. Loose the cork pieces and pull the thread out to increase its length by 10 cm. Now, effective length ~~by~~ 10 cm. Now, length of pendulum becomes 90 cm. Bob will be again 2 cm above the laboratory floor.
21. Repeat steps 13 to 19 two times to take in all the two observations for this new length. Repeat step 20.
22. Repeat step 21 and then steps 13 to 19 to take two observation each for length.

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90 cm, 95 cm, 100 cm, 105 cm and 130 cm.

23. Record all the observations as given ahead.

RESULTS:-

$$\begin{aligned} \text{Experimental length } L &= 100 \text{ cm} \\ \text{Actual length} &= 98.2 \text{ cm} \\ &= 1.8 \text{ cm error} \end{aligned}$$

$$\% \text{ Error} = \frac{1.8}{98.2} \times 100 \approx 1.8\% \quad \boxed{\text{ACTUAL LENGTH WAS ERROR FREE}}$$

This error is within the limit of experimental error.

PRECAUTIONS:-

1. Thread should be strong, weightless and inextensible.
2. Point of suspension should be fixed in a rigid support.
3. Lower faces of split cork should be in same level.

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8. Find the clamp in a stand kept on a table at such a height that the bob just 2 cm above the laboratory floor.
9. Mark a point A on the floor just below the position of bob at rest (mean position).
10. Draw a straight line AB, 10 cm long in direction along which bob will move when oscillating. A is middle point of CB.
11. Find the least count and zero error of the stopwatch. Bring its hand at zero position.
12. Move the bob by hand to over position B ~~or~~ on right of A and leave. See that the bob returns over line BC w/o spinning.
13. When the bob returns from C to A and starts moving to right of A, start the stopwatch and count zero.
14. The bob goes toward B to right extreme, returns from right extreme and goes towards C to left extreme. When bob crosses A from towards C, count one.

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OBSERVATIONS

S.No.	MSR	V.S.D	$D = MSR + V.S.D \times L$
1.	2.5 cm	δ	2.58 cm
2.	2.4 cm	7	2.47 cm

Length and Time

S.No.	$l = l' + x$	t	t_{mean}	$T = \frac{t}{10}$	T_c
1.	l 70 cm	15.3	15.6 $\frac{30.9}{2} \approx 15.5$	1.5s $\frac{15.5}{10}$	2.25 2.25
2.	l 90 cm	18.5	19.1 $\frac{37.6}{2} = 18.8$	1.7s 1.8s	3.24 ≈ 3.25

hook = 1.4 cm
Diameter = 2.5 cm
Radius = 1.25 cm

$$T = \frac{15.5 + 18.8}{2} = \frac{34.3}{2} = 17.15$$

$$T_1 = 15.5/10 = 1.5$$

$$T_2 = 18.8/10 = 1.8$$

CALCULATIONS

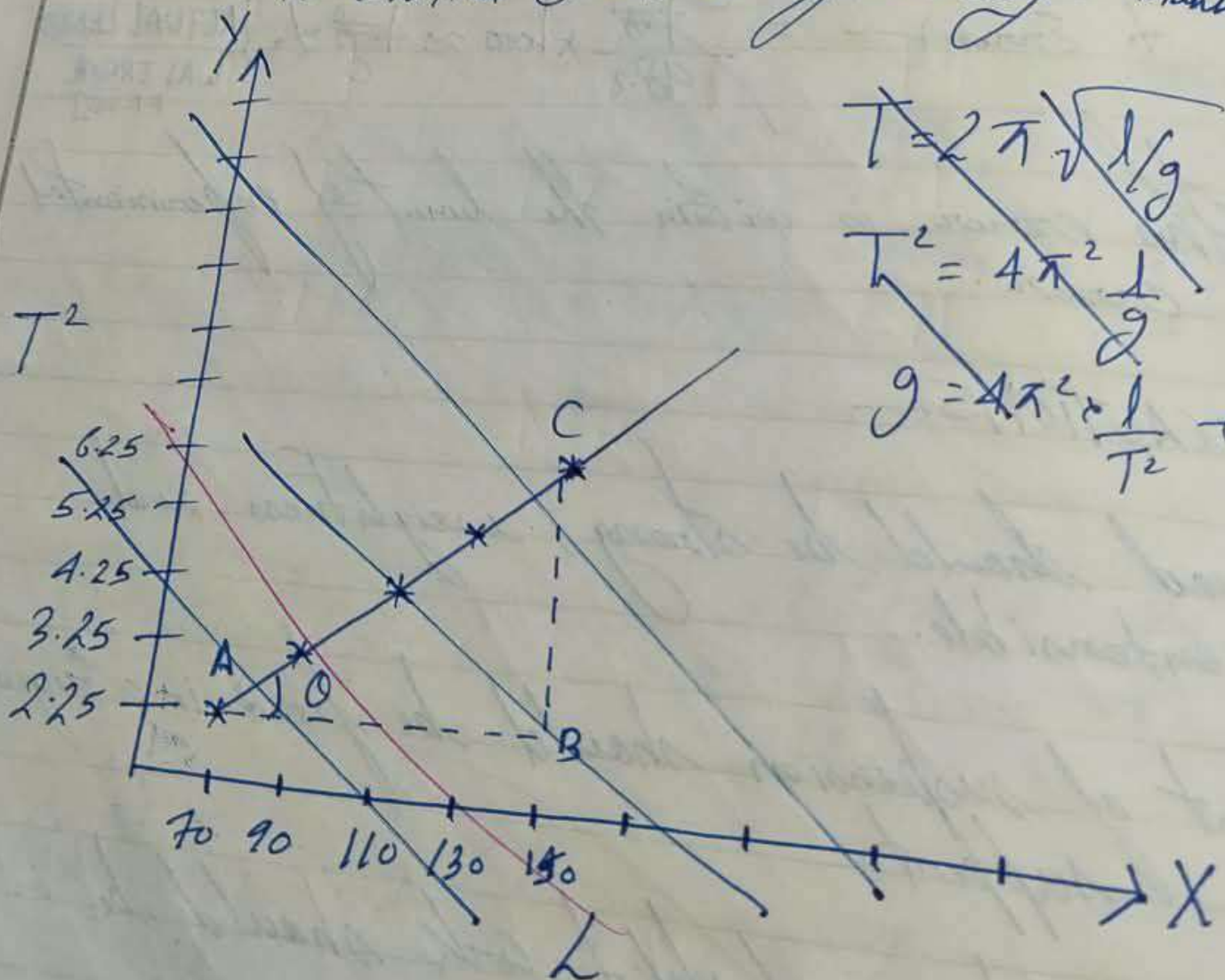
$$D' = \frac{2.5 + \cancel{2.4}}{2} = \cancel{2.45} \quad 2.5$$

$$D = \frac{2.50 - \text{Error}}{2} \quad (\text{Error} = 0)$$

$$= 2.50 \approx 2.5$$

$$l = 1.25 \text{ cm}$$

Note: Actual length was error free, error is written as a formality. Thank You



$$T = 2\pi\sqrt{l/g}$$

$$T^2 = 4\pi^2 \frac{l}{g}$$

$$g = 4\pi^2 \times \frac{l}{T^2} \quad \text{---} \text{ } \odot$$

$$\text{Slope} = \tan \theta = \frac{BC}{AB} = \frac{4}{5}$$

EXTENSION OF THE EXPERIMENT:-

S.No.	l	l'	t	n	$T = t/n$	T^2
1.	70	67.35	31	20	1.55	2.4
2.	90	87.35	37.6	20	1.88	3.5
3.	110	107.35	39.9	20	1.99	3.96 $\times 4$
4.	130	127.35	43.7	20	2.18	4.7
5.	150	147.35	48.2	20	2.41	5.8

NOTE

Data from 3. to 5. have been calculated using home mad lab equipment and verified approx. by an online simulator.

Hence $g = 98.6 \text{ cm/s}^2 \approx 9.8 \text{ m/s}^2$

4. Splitting should be perpendicular to the plane of vibrations of the pendulum.
5. Amplitude should be small to have $\sin \theta = \theta$ [when $\theta \leq 18^\circ$].
6. The bob should move along a straight line.
7. The bob should not spin during vibration.
8. Place of experiment should be free from disturbance of building vibration current.
9. Lab fans should be shut off.
10. Length of pendulum should include length of hook and radius of bob.
11. Counting should be proper and started from zero.
12. Clock/watch should be accurate.
13. Length of pendulum should be increased in steps of 10 cm to bring appreciable change in period.



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Meter scale used should be accurate.

Do corrections

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$$T = 2\pi\sqrt{l/g}$$

$$T^2 = 4\pi^2 l/g$$

$$g = 4\pi^2 \times l/T^2 \rightarrow \theta$$

$$\tan \theta = 0.04$$

Putting θ in (1) :-

$$4\pi^2 \times 25 = \frac{4 \times 9.8596}{25}$$

$$= 98.596 \approx 98$$

Fe

EXPERIMENT-5

AIM: To find the weight of a given body using parallelogram law of vectors.

APPARATUS:

Parallelogram law of forces apparatus (Gauss's apparatus), plumb lines, two hangers with slotted weights, a body (wooden block) whose weight is to be determined, thin strong or thread, white drawing paper sheet, drawing pins, mirror strip, sharp pencil, half meter scale, set squares, protractor.

THEORY:-

If the body of unknown weight S is suspended from middle hanger, known weights P and Q are suspended from other two hangers, then $\vec{P} + \vec{Q} + \vec{S} = 0$

$$S = \sqrt{P^2 + Q^2 + 2PQ \cos \theta} \quad \text{--- (1)}$$

The unknown weight can be calculated from eq. (1).

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PROCEDURE

1. Set up the Gravesand's apparatus with its board vertical, (tested w/ the help of a plumb line).
2. Test that pulleys P₁ and P₂ are frictionless. Oil them if necessary.
3. Pin the white drawing paper sheets on the board w/ the help of drawing pins.
4. Take three pieces of strong thread and tie their one end together to make Knot O. This knot becomes junction of three threads.
5. From the other ends of the two threads, tie a hanger with some slotted weights in each. These serve as the weights P and Q. From the other ends of the third thread tie the given body S.
6. Pass threads with weights P and Q over the pulleys and let the third thread with given body S, stay vertical in the middle of the board.

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CALCULATIONS FOR SLOPE

$$\text{Avg } \bar{l} \Rightarrow \frac{70+90+110+130+150}{5} = 110 \text{ cm}$$

$$\text{Avg } T^2 \Rightarrow \frac{2.4+3.5+4.0+4.7+5.8}{5}$$

$$= 4.08 \text{ cm s}^{-2}$$

l	T^2	$l - \bar{l}$	$T^2 - \bar{T}^2$	$(l - \bar{l})^2$	$(l - \bar{l})(T^2 - \bar{T}^2)$
70	2.4	-40	-1.68	1600	67.2
90	3.5	-20	-0.58	400	11.6
110	4.0	0	-0.08	0	0
130	4.7	20	0.62	400	12.4
150	5.8	40	1.72	1600	68.8

$$\Sigma = 4000, \Sigma = 160.0$$

$$\tan \theta = \frac{\Sigma (l - \bar{l})(T^2 - \bar{T}^2)}{\Sigma (l - \bar{l})^2} = \frac{160}{\frac{4000}{100}} = 0.04 \text{ s}^2/\text{cm}$$

7. Adjust the weight P and Q (forces) such that the junction O stays in equilibrium slightly below the middle of the paper.
8. The weights P, Q and wooden block S acts as three forces P , Q and S acting along the three threads at junction O. The forces are in equilibrium.
9. See that all the weights hang freely and none of them touches the board or the table.
10. Mark the position of junction O on white paper sheet by a sharp pencil.
11. Leave the P and Q as still.
12. Note position of junction O. It must be very close to earlier position.
13. Keeping mirror strip lengthwise under each thread, mark the position of the ends of the image of the thread is mirror, covering the image by the thread (thin mirror, remove parallel error). The positions are P_1 , P_2 and Q_1 for thread of weight P, Q, and Q, for thread of weight Q and S, & for the weight S as shown in figure.

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14. Remove paper from the board.

15. With the help of a half meter scale draw lines through points P_1 and P_2 to represent P , through points Q_1 and Q_2 to represent Q and through points S_1 and S_2 to represent S . These lines must meet at point R .

16. Taking a scale, $1\text{ cm} = 40\text{ g}$, take $OA =$

$OS = 5\text{ cm}$ and $OQ = 5\text{ cm}$ to represent

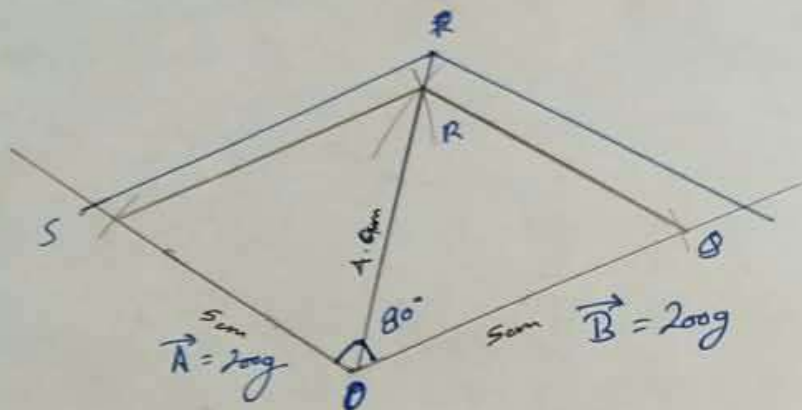
$A = 200\text{ g}$ & $B = 200\text{ g}$

17. Complete $\parallel^m$ $OQRS$ using set squares and join OR . It represents R .

18. Measure OR . It comes to be 4.9 cm

19. Find different sets of observations, change O & Q suitably.

20. Find the weight of wooden block by spring balance.



$$\begin{aligned}
 \vec{R} &= 40 \times 40 = 1600 \\
 \vec{R} &= 4.9 \times 40 = 196 \\
 \vec{R} &= 51 \times 40 = 2040 \\
 \vec{R} &= \frac{1600 + 196 + 2040}{2} \approx 143.6
 \end{aligned}$$

$$\theta = 80^\circ$$

Taken Unit :- $200g = 5cm$
 $1cm = 40g$

CALCULATION + $|\vec{R}| = \sqrt{A^2 + B^2 + 2AB \cos \theta}$

$$= \sqrt{(200)^2 + (200)^2 + 2(200)^2 \cos 80^\circ}$$

$$= 306.4g$$

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RESULT:-

The unknown weight of a given body = 306 g .


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EXPERIMENT-7

AIM: To determine mass of a given body using a meter scale by principle of moments.

APPARATUS: A meter scale, a broad heavy wedge with sharp edge, a weight pan, a body of unknown mass.

THEORY: If m and M be the mass of the body and mass of the weight used and a_1 and a_2 be the distances of their loops from wedge, then, power (mass) arm = a_1 , weight arm = a_2 .

From principle of Moments,

$$mg a_1 = Mg a_2$$

$$m = \frac{M a_2}{a_1}, \text{ which can be calculated}$$

Two Different Methods

i) Arm length fixed and equal, weight adjustable.

The thread loops are suspended at position forming both arms of equal length. Weights in the



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PROCEDURE:-

1. Arrange the meter scale horizontally by supporting it at the sharp edge of the board using wedge at 50cm mark.
2. Suspend the body of the unknown mass by a loop thread at a fixed mark on the left of the wedge.
3. Suspend paper pan at same distance on the right of the wedge with some weights in it.
4. Adjust the weights in paper pan till the meter scale becomes horizontal.
5. Note the mass of the weights in the pan.
6. Repeat steps 2 to 5, three times by increasing the length of the arm in equal steps keeping the lengths equal.
7. Record the observations.

RESULT: The unknown mass of the body, $m = 65.8g$,
 $m = 63.75g$.



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papers pan are adjusted till the meter scale becomes horizontal.

In this case,

$$\text{Hence, } a_1 = a_2 = a$$

$$mga_1 = Mga_2$$

$$m = M$$

A physical balance makes use of this method.

ii) Masses and power arm fixed, weight arm adjustable

Mass is suspended at a fixed distance a , length of power arm is adjustable by moving weight loop thread in and out till the meter scale becomes horizontal.

In this case, $a_1 = a$, $a_2 = A$

$$mga_1 = Mga_2, \text{ becomes } mga = MA$$

$$m = \frac{MA}{a}$$

OBSERVATIONS

S.No	M	H	h	m	CG
1.	100g	16.8cm	26.1cm	64.3g	49.4cm
2.	100g	7.4cm	11.7cm	63.2g	48.3cm

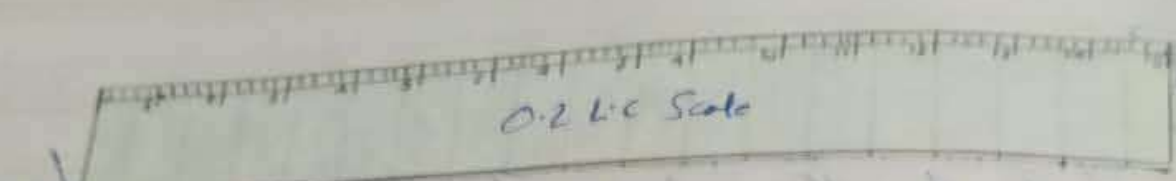
Mean Mass

(CALCULATIONS)

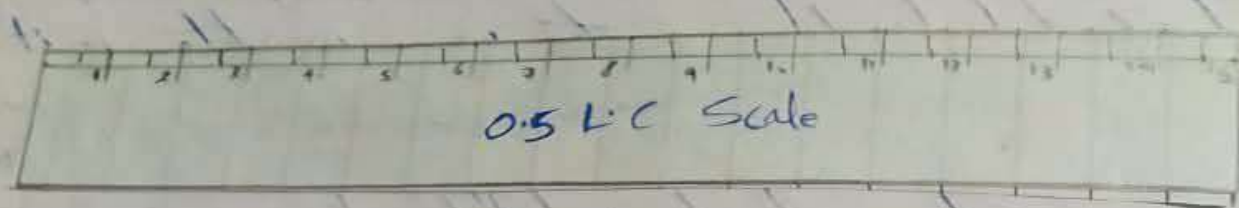
$$\Rightarrow \frac{64.3 + 63.2}{2}$$

$$\approx 63.75g$$

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0.2 L.C. Scale



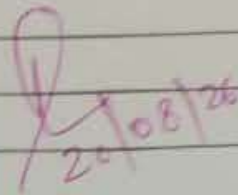
0.5 L.C. Scale

Result: $m = (3.45)$

Join O to L and 2 equal divisions for O to L .

Draw a line RS $\parallel$ to PO at a distance of 2 cm.

Cut the rectangular strip $POPS$ using scissors.


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ACTIVITY - 1 (EXPERIMENT - 5)

AIM:- To make a paper scale of least count 0.2 & 0.5 mm

APPARATUS: a thick white paper sheet, a plastic metre ruler with sharp edge marked in cm and mm, sharp pencil, pair of scissors, journal.

PROCEDURE:-

1. Fold the white Paper along the length.
2. Draw a line P-Q 15 cm along the length. Mark 0 at pt P, mark points on the line PQ at 1 cm mark separation.
3. Draw lines of length 5 mm $\perp$ to the PQ at 10 mm separation, and on mark points and label them as 1, 2, 3, ... to 15.
4. Draw a line AB $\parallel$ to PQ at a distance of 3 mm.
5. Divide each cm division in 5 equal divisions.



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